MILITARY SPECIFICATION

LINE ASSEMBLIES, RADIO FREQUENCY TRANSMISSION GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope. This specification covers the general requirements for assemblies used with standard size, radio frequency transmission lines, for general application by the armed services. The lines are designed to have a nominal impedance of 50 ohms or 75 ohms.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-403	- Plastic Molding Material, Polytetrafluoroethylene (Tetrafluorocarbon).
QQ-B-654	- Brazing Alloy, Silver.
QQ-S-365	- Silver Plating, Electrodeposited; General Requirements for.
ZZ-R-765	- Rubber, Silicone.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.
PPP-T-60	- Tape: Pressure-Sensitive Adhesive, Waterproof, For Packaging.
PPP-T-76	- Tape, Pressure-Sensitive, Adhesive Paper, (For Carton Sealing).

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MIL-P-116	- Preservation, Methods of.
MIL-L-3890	- Lines, Radio Frequency Transmission (Coaxial, Air Dielectric), General Specification for.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-W-6858	- Welding, Resistance, Aluminum, Magnesium, Non-hardening Steels or Alloys, Nickel Alloys, Heat-resisting Alloys, and Titanium Alloys, Spot and Seam.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-S-13282	- Silver and Silver Alloys.
MIL-F-24044	- Flanges, Coaxial Line, Rigid Air Dielectric, General Specification for.
MIL-C-45662	- Calibration System Requirements.
MIL-L-28796/1(NAVY)	- Line Assemblies, Radio Frequency Transmission 1-5/8 Inch Elbows.
MIL-L-28796/2(NAVY)	- Line Assemblies, Radio Frequency Transmission 3-1/8 Inch Elbows.

STANDARDS

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads for 40" x 48" Pallets.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-1285	- Marking of Electrical and Electronic Parts.
AN 6227	- Packing, "O" Ring Hydraulic.

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B-21-66A - Naval Brass Rod, Bar and Shapes.
B-103-71 - Phosphor Bronze Plate, Sheet, Strip and Rolled Bar.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE INC (ANSI)

B46.1-1962 - Surface Texture.
B125.15-1970 - Seamless and Welded Austenitic Stainless Steel Tubing for General Services.
H8.1-1970 - Free-Cutting Brass Rod, Bar and Shapes for Use in Screw Machines.
H33.1-1969 - Copper-Zinc-Lead (Leaded Red Brass or Hardware Bronze Rod, Bars, and Shapes).
H37.1-1970 - Seamless Copper and Copper-Alloy Rectangular Waveguide Tube.
H38.3-1972 - Aluminum-Alloy Drawn Seamless Tubes.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Specification sheets. The individual assembly requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of conflict between the requirements of this specification and the specification sheets, the latter shall govern.

3.2 First article. Assemblies furnished under this specification shall be products which have been tested and have passed the first article inspection specified in 4.5 (see 6.3).

3.3 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the assemblies to meet the performance requirements of this specification. Acceptance or approval of any constituent shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Aluminum. Aluminum shall conform to the chemical requirements for alloy 1100 of ANSI H38.3.

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3.3.1.1 Aluminum alloy. Aluminum alloy shall conform to the chemical composition and tensile property requirements for 6063 or 6061 of ANSI H38.3 as specified (see 3.1). In addition, aluminum assemblies shall be furnished in the "F" temper which shall, upon proper heat treatment and aging, be capable of developing the mechanical properties of the -T6 temper of ANSI H38.3.

3.3.2 Copper. Copper shall conform to type OF or DLP of ANSI H37.1.

3.3.2.1 Copper alloy. Copper alloy shall conform to commercial bronze, 90 percent, of ANSI H37.1.

3.3.3 Silver alloy. Silver alloy shall conform to the chemical composition requirements for grade C (coin silver) of MIL-S-13282. In addition, silver alloy assemblies shall be furnished in the drawn temper with a Rockwell hardness of 45.0 to 80.0 on the "B" scale.

3.3.3.1 Silver-lined copper or copper alloy. Silver-lined copper or copper alloy shall be copper (see 3.3.2) or copper alloy (see 3.3.2.1) laminated with a minimum of 0.005 inch of fine silver in accordance with grade A of MIL-S-13282.

3.3.4 Brass. Brass shall conform to ANSI H33.1, H8.1 or ASTM B21.

3.3.4.1 Leaded red brass. Leaded red brass shall be half-hard, in accordance with ANSI H33.1.

3.3.4.2 Leaded brass. Leaded brass shall be half-hard, in accordance with ANSI H8.1.

3.3.4.3 Naval brass. Naval brass shall be alloy A or C, half-hard, in accordance with ASTM B21.

3.3.5 Gaskets. "O" ring gasket shall conform to ZZ-R-765 or AN 6227 as specified (see 3.1).

3.3.6 Inner connector. The inner connector used with the flanges shall be made of phosphor bronze or free-cutting brass in accordance with ASTM B-103 or ANSI H8.1, as applicable, silver plated in accordance with QQ-S-365 (see 3.1). Contact surfaces shall have a 32-microinch finish before plating unless otherwise specified (see 3.1).

3.3.7 Insulator. The inner connector insulator shall be made of polytetrafluoroethylene conforming to L-P-403.

3.3.8 Locating pin. The locating pin shall be made of corrosion-resistant (stainless) steel, type 303 of ANSI B125.15 (see 3.1).

3.3.9 Bolts, nuts and lockwashers. The bolts, nuts, and lockwashers supplied with flanges shall be made of corrosion-resistant steel, type 303 or 316 in accordance with ANSI B125.15 (see 3.1).

3.3.10 Fungus-inert materials. All materials used in the construction of assemblies shall be in accordance with Requirement 4 of MIL-STD-454.

3.4 Design and construction. Assemblies shall be of the design, construction, and overall physical dimensions specified, and shall be capable of fitting in the specified size of transmission line (see 3.1).

3.4.1 Metal parts. All metal parts shall be free from chips, burrs, and scratches in accordance with good commercial practice. Electrical conducting brass parts shall be silver-plated not less than 0.0003-inch thick and then rhodium flashed. The surfaces of all copper assemblies shall be chemically treated with Iridite 7P or equal. The surfaces of all aluminum assemblies shall be chemically treated in accordance with MIL-C-5541.

3.4.2 Dissimilar metals. Dissimilar metals between which an electromotive couple may exist shall not be placed in contact with each other. See MIL-STD-454, Requirement 16 for definition of dissimilar metals.

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3.4.3 Threaded parts. All threaded parts shall be in accordance with Handbook H28.

3.4.4 Connectors. Connectors shall be as specified (see 3.1).

3.4.5 Transmission lines. Transmission lines used as part of the assembly shall conform to MIL-L-3890, except as otherwise specified (see 3.1).

3.4.6 Coaxial flanges. Coaxial flanges used as part of the assembly shall conform to MIL-F-24044 except as otherwise specified (see 3.1).

3.4.7 Angular alinement. In assembling flanges to an assembly, the flange face shall make an angle with the internal surfaces of the assembly as follows:

(a) 1-5/8 inch outside diameter and smaller - 90 degrees ±15 min
(b) Larger sizes - 90 degrees ±30 min

3.4.8 Brazed or welded joints. When a flange is attached to an assembly by a braze or weld this shall be done in accordance with MIL-B-7883 or MIL-W-6858.

3.4.9 Soldered joints. When a flange is attached to an assembly by soldering, silver solder shall be used and shall conform to QQ-S-654.

3.4.10 Surface roughness. When applicable, the maximum surface roughness for all finished mating and inner surfaces of the assembly shall be as specified (see 3.1); flaws shall be included in the roughness height measurement (see ANSI B46.1).

3.5 RF insertion loss. When assemblies are tested as specified in 4.7.2, the insertion loss shall not exceed that specified (see 3.1).

3.6 Voltage standing wave ratio (VSWR). When assemblies are tested as specified in 4.7.3, the voltage standing wave ratio shall not exceed the value specified (see 3.1).

3.7 Impedance. When assemblies are tested as specified in 4.7.4, the impedance value shall not exceed the value specified (see 3.1).

3.8 Velocity of propagation. When assemblies are tested as specified in 4.7.5, the velocity of propagation value shall not be less than specified (see 3.1).

3.9 RF power handling capability. When assemblies are tested for average power handling capability and also for peak power handling capability as specified in 4.7.8, the insertion loss and VSWR for each shall be as specified (see 3.1).

3.10 Dielectric withstanding voltage (when specified, see 3.1). When assemblies are tested as specified in 4.7.6, there shall be no evidence of voltage breakdown.

3.11 Pressurization (when specified, see 3.1). When assemblies are tested as specified in 4.7.7, there shall be no loss of pressurization, as indicated by bubbles.

3.12 Vibration, high frequency. When assemblies are tested as specified in 4.7.9, the insertion loss and VSWR shall be as specified (see 3.1).

3.13 Salt-spray. When assemblies are tested as specified in 4.7.10, dielectric withstanding voltage, the insertion loss, and VSWR shall be as specified (see 3.1).

3.14 Marking. Assemblies shall be marked in accordance with MIL-STD-1285 on one side. The manufacturer's name or trademark may also be marked on the assembly, provided such letters are not expressly forbidden in the contract or order, and their height does not exceed 0.5 inch. The markings shall be applied with permanent ink or paint. All such markings shall withstand normal abuse and be legible for the life of the assembly.

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3.15 Workmanship. Assemblies shall be processed in such a manner as to be uniform in quality and shall be free from pits, corrosion, cracks, rough edges, chips, burrs, scratches, flaws and other defects that will affect life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Materials inspection (see 4.4).
(b) First article inspection (see 4.5).
(c) Quality conformance inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.4 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table I, used in fabricating the assembly are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

TABLE I. Materials inspection.

Material	Requirement paragraph	Applicable document
Aluminum	3.3.1	ANSI H38.3
Aluminum alloy	3.3.1.1	ANSI H38.3
Copper	3.3.2	ANSI H37.1
Copper alloy	3.3.2.1	ANSI H37.1
Silver alloy	3.3.3	MIL-S-13282
Silver-lined copper or copper alloy	3.3.3.1	MIL-S-13282
Brass	3.3.4	ANSI H33.1, H8.1 or ASTM B21
Rubber	3.3.5	AN 6227 or ZZ-R-765
Bronze	3.3.6	B-103
Polytetrafluoroethylene	3.3.7	L-P-403
Corrosion-resistant steel	3.3.8 and 3.3.9	ANSI B125.15
Fungus-inert	3.3.10	MIL-STD-454

4.5 First article inspection. First article inspection shall be performed by the supplier, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract or purchase order under which it is granted.

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4.5.1 Sample. Six assemblies shall be subjected to first article inspection.

4.5.2 Inspection routine. Sample units shall be subjected to first article inspection specified in table II in the order shown. All sample units shall be subjected to the inspection of group I. The sample units shall then be divided equally into three groups of two sample units each and subjected to the inspection for their particular group.

TABLE II. First article inspection.

Examination or Test	Requirement paragraph	Method paragraph
Group I (all sample units)		
Visual and mechanical examination	3.1, 3.3 thru 3.4 incl, 3.14 and 3.15	4.7.1
RF insertion loss	3.5	4.7.2
VSWR	3.6	4.7.3
Impedance	3.7	4.7.4
Velocity of propagation	3.8	4.7.5
Dielectric withstanding voltage (when specified)	3.10	4.7.6
Pressurization (when specified)	3.11	4.7.7
Group II (two sample units)		
RF power handling capability	3.9	4.7.8
RF insertion loss	3.5	4.7.2
VSWR	3.6	4.7.3
Group III (two sample units)		
Vibration, high frequency	3.12	4.7.9
RF insertion loss	3.5	4.7.2
VSWR	3.6	4.7.3
Group IV (two sample units)		
Salt spray (corrosion)	3.13	4.7.10
Dielectric withstanding voltage (when specified)	3.10	4.7.6
RF insertion loss	3.5	4.7.2
VSWR	3.6	4.7.3

4.5.3 Failures. One or more failures shall be cause for refusal to grant first article approval.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection.

4.6.1.1 Inspection lot. An inspection lot shall consist of all the assemblies of the same part number, produced under essentially the same conditions and offered for inspection at one time.

4.6.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table III and shall be made on the same set of sample units, in the order shown.

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TABLE III. Group A inspection.

Examination or Test	Requirement paragraph	Method paragraph	AQL percent defective	
			Major	Minor
Visual and mechanical examination	3.1, 3.3 thru 3.4 incl 3.14 and 3.15	4.7.1	1.0	4.0
RF insertion loss	3.5	4.7.2	1.0	
VSWR	3.6	4.7.3	1.0	
Impedance	3.7	4.7.4	1.0	

4.6.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality levels (AQL) shall be as specified in table III. Major and minor defects shall be as defined in MIL-STD-105.

4.6.1.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.6.1.3 Group B inspection. Group B inspection shall consist of the tests specified in table IV, in the order shown, and shall be made on sample units which have been subjected to and have passed the group A inspection.

TABLE IV. Group B inspection.

Test	Requirement paragraph	Method paragraph
Velocity of propagation	3.8	4.7.5
Dielectric withstanding voltage (when specified)	3.10	4.7.6
Pressurization (when specified)	3.11	4.7.7

4.6.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for special inspection level S-4. The sample size shall be based on the inspection lot size from which the sample was selected for group A inspection. The AQL shall be 6.5 percent defective.

4.6.1.3.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.6.1.3.3 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered on the contract or purchase order, if the lot is accepted.

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4.6.2 Inspection of preparation for delivery. The sampling and inspection of the preservation-packaging and interior package marking shall be in accordance with the groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.7 Method of examination and test.

4.7.1 Visual and mechanical examination (see 3.1, 3.3 through 3.4 incl., 3.14 and 3.15). Assemblies shall be examined to verify that the material, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements.
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4.7.2 RF insertion loss (see 3.5). The insertion loss of the assembly shall be measured over the specified frequency range (see 3.1). The test equipment(s) shall be capable of providing a continuous measurement of insertion loss over the required frequency ranges. A means shall be provided for producing a permanent record of the assembly's insertion loss versus frequency. The measurement system and permanent record shall be calibrated and shall be capable of differentiating an insertion loss change of 0.05 dB.

4.7.3 VSWR (see 3.6). The VSWR of the assembly shall be measured over the specified frequency range (see 3.1). The test equipment(s) shall be capable of providing a continuous measurement of VSWR over the required frequency ranges. A means shall be provided for producing a permanent record of the assembly's VSWR versus frequency. If VSWR is not directly measured; that is, if return loss is measured and VSWR is calculated from that measurement, the permanent record shall indicate the worst case VSWR numerically for each frequency band and shall provide the calculation used to obtain the calculated VSWR. The measurement system and permanent record shall be able to differentiate a VSWR change of at least 0.02.

4.7.4 Impedance (see 3.7). The characteristic impedance of the assembly shall be measured with Time Domain Reflectometry (TDR) equipment. The TDR equipment shall have a total system response of 150 pico-seconds and shall provide for a resolution of 0.1 ohm throughout the length of the assembly. The test results shall be displayed on calibrated rectangular coordinates and a permanent record of the test results shall be produced. The measurement system and permanent record shall be capable of differentiating an impedance difference of at least 0.1 ohm in 50 ohms and 0.15 ohm in 75 ohms. A section of air line with a calibrated impedance of 50.0 $\pm$0.1 ohms or 75.0 $\pm$0.15 ohms is required as a reference.

4.7.5 Velocity of propagation (see 3.8). The velocity of propagation of the assembly shall be determined as follows:

4.7.5.1 Test equipment: The following test equipment shall be utilized:

(a) A microwave sweep oscillator with a frequency range that covers the specified frequency range of the assembly.
(b) An analyzer with suitable readout features.
(c) A phase bridge with a measurement accuracy of $\pm$0.1 percent.
(d) A calibrated section of air line whose length equals the length of the assembly.

4.7.5.2 Test procedures:

(a) Calibrate the system over the specified frequency range of the assembly with required adapters inserted in the leg of the phase bridge to be used for the assembly. Record calibration line.
(b) Insert the assembly in the appropriate leg of the phase bridge and an equal length of air line in the other leg. Note the length (L) of the assembly.
(c) Using the phase bridge, adjust the electrical length of the air line leg to obtain a phase balance across the specified frequency of the assembly. Note the added length (ΔL).

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(d) Calculate velocity of propagation (Vp) of calibration line as follows:

$$Vp = \frac{L}{L + \Delta L} \times 100\%$$

(e) Sweep over the specified frequency of the assembly. Note maximum deviation (ø in degrees) from calibration line.

(f) Calculate minimum velocity of propagation Vp (Min) as follows:

$$Vp\ (Min) = \frac{L}{L + \Delta L + \frac{ø\ C}{360\ f}}$$

where:
C is the velocity of propagation in free space.
f is the frequency (in Hz) at which ø was measured.

4.7.6 Dielectric withstanding voltage (when specified) (see 3.10). The dielectric withstanding voltage shall be measured in accordance with MIL-STD-202, method 301. The following details shall apply:

(a) Magnitude of test voltage:

(1) As specified (see 3.1).

(b) Nature of potential - AC rms.

(c) Points of application of test voltage:

(1) Between the center conductor and the outer conductor.

4.7.7 Pressurization (when specified) (see 3.11). With the assembly ends properly sealed, the assembly shall be subjected to the internal gas pressure specified (see 3.1) for at least 20 seconds while immersed in tap water at a temperature of approximately 20°C. Unless otherwise specified (see 3.1), the internal gas pressure shall be 30 ±1 pound per square inch. During immersion, the assembly shall be observed for evidence of leakage as indicated by escaping air bubbles.

4.7.8 RF Power handling capability (see 3.9). The power handling capability shall be determined by subjecting the assemblies to average RF power at the specified lowest frequency and the highest temperature continuously for 30 minutes or until the assembly temperature stabilizes whichever comes first. The assemblies shall also be subjected to peak power at the specified lowest frequency and the highest temperature for 30 minutes or until the assembly temperature stablizes whichever comes first. The assembly shall be terminated in a match load. Following each power test, RF insertion loss and VSWR shall be measured as specified (see 4.7.2 and 4.7.3).

4.7.9 Vibration, high frequency (see 3.12) The assembly shall be tested in accordance with method 204 of MIL-STD-202. The following details shall apply:

(a) Method of mounting - Attach assembly to the table by a clamp. No part of the assembly shall touch any object other than the clamp.

(b) Test condition letter - B.

(c) Examination after test - RF insertion loss and VSWR shall be measured as specified in 4.7.2 and 4.7.3.

4.7.10 Salt spray (corrosion) (see 3.13). The assembly shall be tested in accordance with method 101 of MIL-STD-202. The assembly ends shall not be covered during test. The following details shall apply:

(a) Test condition letter - A.

(b) Measurements after exposure - Dielectric withstanding voltage, RF insertion loss, and VSWR shall be measured as specified in 4.7.6, 4.7.2 and 4.7.3.

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5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual orders.)

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Line assemblies shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Line assemblies shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Line assemblies shall be individually packaged in accordance with MIL-P-116, submethod IIe insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein. The container shall conform to PPP-B-566, PPP-B-676 or PPP-B-636.

5.1.1.5 Intermediate packaging. Not required.

5.1.2 Level C. Line assemblies shall be clean, dry and packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. The packaged line assemblies shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirement in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using non-metallic or tape banding only.

5.2.2 Level B. The packaged line assemblies shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. The packaged line assemblies shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Line assemblies, packed as specified in 5.2.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.2.4.2 Level B. Line assemblies, packed as specified in 5.2.2, shall be unitized as specified in 5.2.4.1 except that the fiberboard caps shall be class domestic.

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5.2.4.3 Level C. Line assemblies, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size and kind commonly used for the purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), each unit package, exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General. Exterior containers (see 5.2.1, 5.2.2 and 5.2.3) shall be of minimum tare and cube consistant with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

6. NOTES

6.1 Intended use. Transmission line assemblies are intended for operation in the microwave region of the spectrum.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet and the military part number (see 3.1).
(c) Levels of preservation-packaging and packing required (see 5.1 and 5.2).
(d) Special marking, if required (see 5.3).

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Review activities:

Navy- SH, OS, AS

User activities:

Navy - MC

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-N393)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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